

This assignment will not be graded and is only for practice.

Key Points: 1

Let $T : V \rightarrow W$ be a linear transformation. We define kernel of T (denoted by $\ker(T)$) and image of T (denoted by $\text{Im}(T)$) as follows:

$$\ker(T) = \{v \in V \mid T(v) = 0\}$$

$$\text{Im}(T) = \{w \in W \mid \text{there exists } v \in V \text{ for which } T(v) = w\}$$

T is injective (i.e., T is a monomorphism) if and only if $\ker(T) = \{0\}$.

A linear transformation $T : V \rightarrow W$ is surjective (i.e., T is epimorphism) if and only if $\text{Im}(T) = W$.

$\text{Nullity}(T) = \dim(\ker(T))$ and $\text{Rank}(T) = \dim(\text{Im}(T))$.

Let $T : V \rightarrow W$ be a linear transformation. Then

$$\text{Rank}(T) + \text{Nullity}(T) = \dim(V).$$

This is known as the rank nullity theorem.

Let $T : V \rightarrow W$ be a linear transformation. T is injective if and only if $\text{Nullity}(T) = 0$. T is surjective if and only if $\text{Rank}(T) = \dim(W)$.

Consider the following linear transformation:

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

$$T(x, y, z) = (2x + 3z, 4y + z)$$

Answer the following questions:

1) Which of the following Options is true?

1 point

[Hint: $T(x, y, z) = (2x + 3z, 4y + z) = (0, 0)$, i.e., $2x + 3z = 0$ and $4y + z = 0$]

- ☐ $\ker(T) = \{(3y, y, -4y) \mid y \in \mathbb{R}\}$
- ☐ $\ker(T) = \{(6y, y, 4y) \mid y \in \mathbb{R}\}$
- ☐ $\ker(T) = \{(6y, y, -4y) \mid y \in \mathbb{R}\}$
- ☐ $\ker(T) = \{(3y, y, 4y) \mid y \in \mathbb{R}\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\ker(T) = \{(6y, y, -4y) \mid y \in \mathbb{R}\}$$

2) Which of the following is a basis of $\ker(T)$?

1 point

- ☐ $\{(3, 1, -4)\}$
- ☐ $\{(6, 0, -4), (0, 1, -4)\}$
- ☐ $\{(6, 1, -4)\}$
- ☐ $\{(3, 0, -2), (0, 1, -4)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\{(6, 1, -4)\}$$

3) Find the nullity of T .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

4) Find the rank of T .

[Hint: use rank nullity theorem.]

No, the answer is incorrect.

Score: 0

Feedback:

– Here $\dim(V) = 3$ as $V = \mathbb{R}^3$.

Accepted Answers:

(Type: Numeric) 2

1 point

5) Which of the following is true?

1 point

- ☐ T is both injective and surjective.
- ☐ T is injective but not surjective.
- ☐ T is surjective but not injective.
- ☐ T is neither injective nor surjective.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is surjective but not injective.

6) Which option represents the kernel and image of the following linear transformation?

1 point

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(x, y) = (x, 0)$$

- ☐ $\ker(T) = \text{Span}\{(1, 0)\}, \text{Im}(T) = \text{Span}\{(1, 0)\}.$
- ☐ $\ker(T) = \text{Span}\{(1, 0)\}, \text{Im}(T) = \text{Span}\{(0, 1)\}.$
- ☐ $\ker(T) = \text{Span}\{(0, 1)\}, \text{Im}(T) = \text{Span}\{(1, 0)\}.$
- ☐ $\ker(T) = \text{Span}\{(0, 1)\}, \text{Im}(T) = \text{Span}\{(0, 1)\}.$

No, the answer is incorrect.

Score: 0

Feedback:

• $\text{Ker}(T) = \{(0, y) \mid y \in \mathbb{R}\}.$

Accepted Answers:

$\ker(T) = \text{Span}\{(0, 1)\}, \text{Im}(T) = \text{Span}\{(1, 0)\}.$

Key Points: 2

Let $T : V \rightarrow W$ be a linear. Follow the steps below to find bases for the null space and the range space of T .

Step 1: Find the matrix A corresponding to T with respect to some standard ordered bases $\beta = \{v_1, v_2, \dots, v_n\}$ and $\gamma = \{w_1, w_2, \dots, w_m\}$ for V and W respectively.

Step 2: Use row reduction on A to obtain the matrix R which is in reduced row echelon form.

Step 3: The basis of the solution space of $Rx = 0$ is the basis of null space of matrix A and can be obtained by finding the pivot and non-pivot columns (dependent and independent variables) as seen earlier.

Step 4: The vectors $\begin{bmatrix} c_{11} \\ c_{12} \\ \cdot \\ \cdot \\ c_{1n} \end{bmatrix}, \begin{bmatrix} c_{21} \\ c_{22} \\ \cdot \\ \cdot \\ c_{2n} \end{bmatrix}, \dots, \begin{bmatrix} c_{k1} \\ c_{k2} \\ \cdot \\ \cdot \\ c_{kn} \end{bmatrix}$ form a basis of the null space of A precisely when

the vectors $v'_1, v'_2, \dots, v'_k \in \ker(T)$ where $v'_i = \sum_{j=1}^n c_{ij} v_j$, form a basis for $\ker(T)$. Use the basis obtained in step 3 to thus get a basis for $\ker(T)$.

Step 5: Recall that if $i_1, i_2, \dots, i_r$ are the columns of R containing the pivot elements, then the same columns of A form a basis for the column space of A .

Step 6: The vectors $\begin{bmatrix} d_{11} \\ d_{12} \\ \cdot \\ \cdot \\ d_{1m} \end{bmatrix}, \begin{bmatrix} d_{21} \\ d_{22} \\ \cdot \\ \cdot \\ d_{2m} \end{bmatrix}, \dots, \begin{bmatrix} d_{r1} \\ d_{r2} \\ \cdot \\ \cdot \\ d_{rm} \end{bmatrix}$ form a basis of the column space of A precisely

when the vectors $w'_1, w'_2, \dots, w'_r \in \text{im}(T)$ where $w'_i = \sum_{j=1}^m d_{ij} w_j$, form a basis for $\text{im}(T)$. Use the basis obtained in step 5 to thus get a basis for $\text{im}(T)$.

Consider the following linear transformation:

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T(x, y, z) = (x - z, 2x + 3y + z, 3y + 3z)$$

Answer the following questions 7-9 using the information given above.

7) Which of the matrices corresponds the given linear transformation T with respect to the standard ordered basis of $\mathbb{R}^3$ for both the domain and the codomain? **1 point**

☐ $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 3 \\ -1 & 1 & 3 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 2 & 3 \\ -1 & 3 & 3 \\ 0 & 1 & 0 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 1 \\ 3 & 3 & 0 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 3 & 3 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 3 & 3 \end{bmatrix}$$

8) What will be the kernel of T ?

1 point

☐ $\{(x, 0, z), (0, y, z) \mid x, y, z \in \mathbb{R}\}.$

☐ $\{(z, 0, z), (0, -z, z) \mid z \in \mathbb{R}\}.$

☐ $\{(z, -z, z) \mid z \in \mathbb{R}\}.$

☐ $\{(x, 0, z), (0, -x, z) \mid x, z \in \mathbb{R}\}.$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(z, -z, z) \mid z \in \mathbb{R}\}.$

9) What will be the basis of the image space of T found by the algorithm mentioned in **1 point** the Key Points ?

☐ $\{(1, 0, -1), (2, 3, 1)\}$

☐ $\{(1, 0, 0), (0, 1, 0)\}$

☐ $\{(1, 2, 0), (0, 3, 3)\}$

☐ $\{(-1, 1, 3)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1, 2, 0), (0, 3, 3)\}$

Your score is: 0/9